

VirtualVehicle.slx DAE system (Simscape Drive-line)

Differential equation

$$m\dot{v} = F_t - b_f v - \frac{1}{2}\rho C_d A v^2 - mg \sin \theta$$

- $m = 1300$ kg, $b_f = 250$ N · s/m, $\rho = 1.225$, $C_d = 0.6$, $A = 6$ m²
- $F_t = \text{MaxF} \cdot \tanh(u/\text{MaxF})$, $\text{MaxF} = 14800$ N
- $\text{lim} = 0$ (no braking) or $-\infty$ (braking allowed)

Road grade

$$\theta = \arctan\left(\frac{y_{\text{front}} - y_{\text{rear}}}{2.18}\right)$$

$$y_{\text{front}} = \text{pchip}(x_c, y_c, x), \quad y_{\text{rear}} = \text{pchip}(x_c, y_c, x - 2.18)$$

Road profile is generated from spline control points depending on disturbance type (none / hill / bumps).

Comparison with simple model

The VehicleModel.mlx lab starts with the linear first-principles model $m\ddot{x} = -b\dot{x} + u$ ($m = 1000$, $b = 150$) and has students identify effective linear parameters (m , b) from the virtual vehicle's step response. The transfer function matches well for small inputs but diverges for large inputs because the real Simscape model includes: - quadratic aerodynamic drag ($\propto v^2$) - smooth saturation of traction force ($\tanh$) - road grade disturbance ($\sin \theta$)